E-tutor: Comprehensive Student Productivity Management System for Education

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***Abstract*— With the advancement of technology, e-learning has emerged as predominant in the education sector. As students, parents, and educators acknowledged, adopting e-learning can offer several benefits over traditional learning techniques. Since more individuals are becoming acclimated to online learning platforms, these online platforms can provide a simple, instructive, and efficient mode of delivery. This novel approach could be improved with the aid of Artificial Intelligence (AI) to comprehend consumers more thoroughly and provide valuable and better-suited services. Most sectors in education, including universities, swiftly adapted to new educational methodologies because of their flexibility and productivity. Nevertheless, there are some downsides that young demography experiences, such as less instructiveness, distraction due to the absence of teachers, and poor IT literacy. Consequently, these drawbacks would recede the capability of students to assimilate content during the lecture. Therefore, the main objective of this research is to implement an E-learning platform with AI learning analytics to enhance students' performance regularly while reducing the significant drawbacks of the E-learning platforms. This research consists of students' focus detection, essay-based answer evaluation, note summarization, mind map generation, and personalized guidance facilities.**

Keywords—e-learning, artificial intelligence, question generation, personalized guidance, user recommendations, summary generation, mind map generation, object detection

I. INTRODUCTION

E-learning has emerged as one of the most widely used learning techniques of the 21st century because of the apparent rapid evolution of web technology and advanced technologies. A recent pandemic, COVID-19, has significantly influenced educational facilities worldwide [1]. As a result, traditional teaching-learning techniques were replaced with novel approaches [1], [2]. It is a tedious task to evaluate subjective answers manually for technical subjects. Since the same inference procedure is used for all students, uniformity in marking is ensured when evaluating computers utilizing intelligent techniques. The effectiveness and quality of e-learning systems are regularly enhanced by the development of new tools and technology. The survey was developed and distributed to students in secondary-level education to analyze the workload of science subjects and their assessments to ensure the workload per head. Consequently, the majority accepted the difficulty of the traditional process. This fact leads us to conclude that G.C.E. Ordinary Level Science students will benefit greatly from a comprehensive student productive management system to enhance effectiveness and user-friendliness immensely. Therefore, designing this platform more supportive for science students to accomplish their learning activities was the purpose. In addition to that, essay-based evaluation is one of the unique features it expedites the marking of essay-based answers of students. It will also briefly explain the important content the student has missed and the proper way of answering the essay questions. The sophisticated question-evaluation system is based on the student's field of study (e.g.: -science) and the focus detection evaluates the focus level of the students. Moreover, the personalized guidance unit regularly enhances the academic performance levels of the students by providing AI-based questions. Furthermore, the system generates a mind map as a summarized version of the lecture using semantic web technologies.

After the introduction section, the rest of the paper consists of four sections. Section 1, the Literature Survey section evaluates the related contribution to the enhancement of online platforms. Section 2, the Methodology section describes the comprehensive development and techniques involvements. Section 3, the Results and Discussion section illustrates the outcome of the developed system while demonstrating the advantages of applied technology concepts. And Section 4, the Conclusion section discusses the future development.

II. LITERATURE SURVEY

There are two modes of e-learning techniques: synchronous and asynchronous. The synchronous mode provides a real-time teaching-learning experience, and the asynchronous mode enables sharing of knowledge among multiple people not in real-time but during occasions such as performing assessments [3]. Asynchronous mode is preferred by 69.9% of students, according to Georgia State University research conducted during the COVID-19 pandemic through the distance learning approach [4]. Due to the nature of less interaction, it is challenging to keep the focus of students in

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asynchronous mode [5]. When drowsiness can enhance students' unproductive behaviors, students eventually will get bored while studying. Academic achievement is observed to be negatively correlated with student distraction and unproductive behavior [6], [7]. Therefore, achieving an efficient e-learning experience that aids students to retain their learning and determining their current knowledge standard, is essential to increasing productivity [8]. Numerous significant types of research were conducted surrounding distance learning to maximize the enthusiasm of students. Among the research, Krishnan et al. [9] identified that by using the video camera, the student's engagement and productivity can be examined while detecting facial expressions and drowsiness. According to Narayanan et al. [10], user engagement can be revealed from head and hand posture information. Another research by Ashwin T.S. and Ram M.R.G suggested that students' body postures, hand gestures, and facial expressions are important indicators of their emotional moods [11]. There are several ways to enhance engagement and productivity in an asynchronous e-learning environment. Improving essay-based question evaluation and multiple-choice questions (MCQ) evaluation are two basic components that guide students regularly with better performance. Automated Arabic Essay Scoring (AAES) uses Value stream mapping (VSM) and Latent semantic indexing (LSI) to automatically score essay responses using two basic strategies for gathering pertinent data [12],[13]. Researchers [14] developed quiz-based assignments based on Gagne's nine events of instruction to enhance encoding and verification to enhance academic achievement. Another research conducted by Kolodny & Bayly, 1983 used pre-stored questions in the database and prompts randomly [15] as another approach. Comparatively the synchronous methods outweigh the asynchronous mode of educational research.

III. METHODOLOGY

The overall system consists of 4 central units: focus detection, essay-based question evaluation, note-taking and mind map generation, and a personalized guidance unit. The focus detection in the system can gather the students' focus level in real time using camera input and estimate the focus level while providing feedback information about the current session focus level to the users. The essay-based question evaluation unit will give students to evaluate and understand the missing points on their answer sheets. The notetaking unit has two subcomponents: summarization and mind map generation. Note-taking is used for students to take notes efficiently to learn productively. At the same time, summarization and the mind map generation unit will enhance the learning process by summarizing the lectures and generating mind maps for the relevant lectures provided by the students. The student's academic performance level will be handled by the personalized guidance unit. Initially, the system will offer pass papers to the students and analyze the answers. After the system categorizes students based on their answers, the system will generate similar questions based on the performance of answers given to previous papers. Essentially students' performance levels will increase. The personalized guidance unit consists of multiple processes to deliver full performance. The following procedures are analyzing questions provided to students, evaluating the answers, understanding the knowledge level, categorizing, designing question patterns, and generating similar questions based on categorization. Providing questions based on categorization will regularly enhance academic performance levels.

This system architecture is presented in Fig. 1. The system components are independent and will decrease the fault tolerance of the system. Apart from that, the primary users of the system are students. With this structure, the components will intelligently understand the users and enhance the users' performance level regularly in a productive manner.

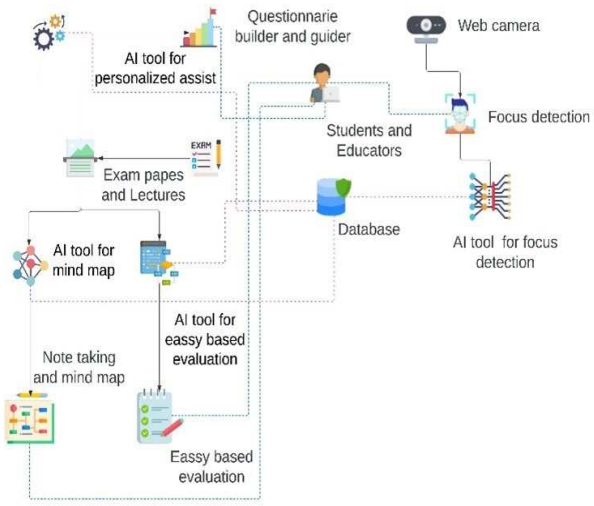


Fig. 1: System architecture

A. Focus Detection

Electronic devices are required for online learning, which can readily divert attention for lengthy periods. Most students will be distracted while they are studying. Even though students may feel convinced that they studied effectively, the fact is that instead of concentrating, they were distracted or studying while drowse. Nevertheless, there is no reliable method to calculate the students' degree of concentration while they are studying. In a classroom, the instructor may encourage the students who have been distracted to refocus. That's because of school structure, and it's one of the most common reasons why online students struggle while learning on their own. Although distractions in the classroom generally exist in the shape of friends and other students, there are several diversions at home, such as studying while exhausted or skipping lectures. It might be difficult for students to concentrate without any advice. A significant portion of the students who use e-learning technologies may get distracted or frustrated. Instructors, guardians, and students will be able to recognize their students' distractions and focus on e-Learning by offering a system that identifies distractions, facial emotions, and expressions in real time. Instructors could even encourage students to pay attention during their future e-learning sessions. Through the web camera, the subsystem will capture the users' (students) facial expressions. Following this section, the neural network will be fed with data from users' facial expressions then the neural network is able to predict users' facial expressions with the help of a trained deep learning model which is hosted on the cloud object storage. Then the predicted facial expressions will be displayed to the

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users. To accomplish the focus detection component, there are six main approaches.

1. Setup TensorFlow Object Detection API pipeline
2. Use transfer learning to train a deep learning model
3. Convert focus detection model to Tensorflow.JS
4. Host the trained TensorFlow deep learning model for applications
5. Create React and Tensorflow.JS Computer Vision Template
6. Detect real-time custom detections using a deployed Tensorflow.JS model and visualize the detections within the HTML canvas

B. Object Detection API and Data Acquisition

Python and OpenCV will be used to retrieve the dataset's custom visuals. In addition, the Kaggle data and custom datasets used to gather necessary facial expression. Then, using a label image package, those photos will be labeled [16]. After that LXML library files which contain the details of labeled portions of the images will be created then it will be divided into training data and testing data in the ratio of an 80:20 split. After the splitting, the pipeline for the TensorFlow object detection API will be created. Then the labeled files will be converted to TensorFlow records. TensorFlow provides an efficient way to construct, train, and deploy object detection models, it is possible to build (pre-trained) object identification models with this [17]. Apart from that, the TensorFlow model zoo, which is part of the TensorFlow object detection API, will be useful when initializing custom focus detection models and training on novel datasets. TensorFlow model zoo contains pre-trained models. It contains the number of pre-trained models that were developed using a variety of datasets, including the Open Images Dataset, the Karlsruhe Institute of Technology and Toyota Technological Institute (KITTI) dataset, and the Common Objects in Context (COCO) dataset.

C. Single Shot Multi-Box Detector (SSD)

Considering the speed and the mean average precision, one of the balanced TensorFlow model zoo will be used to train the model which is SSD MobileNet V2 FPNLite 320x320 with a speed of 22ms and 22.2 COCO map [18]. The mean average precision (map) is a measure of bounding box detection accuracy and recall [19]. It's a useful indicator of both the network's sensitivity to items of interest and its efficiency at minimizing false alarms. The accuracy of the network increases with the map score, but at the expense of execution speed, which needs to be avoided. Low-end devices (laptops and mobile devices) that have limited processing power are required to select the proper mode zoo with less execution time. Also selecting a reasonable map score will increase the performance level. The initial class values are 90 and that has been changed to 4 which is the number of emotions that the model will be trained on the custom focus detection. The hyperparameter used in the training process is the number of steps which is the steps that the mode will train on and, it was set up to 10000 steps. Additionally, the transfer learning technique is used to train the customized focus detection module more quickly. To train a neural network from scratch, a lot of data is required. Transfer learning models, on the other hand, concentrate on transferring the knowledge acquired while addressing one problem to another that is related, therefore they may be developed with very minimal training data because the model has already been pre-trained [20]. Additionally, the transfer learning technique can gain better performance of neural networks with less training period. After the train sections are finished, the pre-existing TensorFlow Python model will be translated into a TensorFlow JS (TFJS) compatible version and hosted on cloud object storage, enabling the web application to access the model from anywhere in the world. A derivative part of the system is to calculate the users'(students) distraction levels and send frequent occurrences of the distractions to the guardians of the students. The system will compile the frequent facial expressions and occurrences over a certain period and send that data to the cloud database to relay the distraction levels. The system will determine at the end of the month if the frequent occurrences are productive (happy, neutral) or not (bored and drowsy). The system will then compile the ineffective behaviors and facial expressions that frequently occurred in the preceding month and transmit this data to the students' guardians.

D. Essay-Based Question Evaluation

Comparing the students' responses to the lecturer's answer sheet is a process handled by the essay-based question evaluation unit. In this comparison, the semantic similarity concept is used for detecting text similarity. The similarities in semantic vectors are taken into consideration when evaluating essay answers. It is essential to examine keywords while evaluating essay answers. As a result, the sklearn library and TfidfVectorizer function are used to perform term weighing. The student and the sample answers are considered two separate documents when creating the term frequency-inverse document frequency (Tf-idf) for term weighing. Before being vectorized, the text is pre-processed by eliminating special characters such as stopping words and altering it to the lowercase. After vectorizing, cosine similarity is used to calculate pairwise similarities of the tf-idf vectors. Answers to essay-type questions should be between six and ten sentences long. The DistilBERT model determines the semantic similarity of essay-type responses. Above pre-trained Google-developed transformer model is smaller, faster, affordable, and light. It is based on the Bidirectional Encoder Representations from Transformer (BERT) architecture model. DistilBERT is trained using unlabeled text from sources such as the 16GB of Wikipedia and the Books Corpus that were used in BERT. Pre-trained DistilBERT model is improved for this semantic similarity and fine-tuned by training it with science past papers and answers. To train the model, the scores are standardized to a range of 0-100. After the model has been given the students' answer and sample answer pairs, it produces text vectors. Cosine similarity is used to determine how similar the embedded text vectors are. Then, semantic similarity is predicted using the degree of similarity between these representations. The review procedure is more comprehensive when ontologies are used since it is feasible to check for the presence of keywords, synonyms, the appropriate word combination, and concept coverage. The above methods are put into practice using ontologies and assessed using common input data made up of technical

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answers of science. The analysis of the results leads to the conclusion that ontology provides more accurate results.

E. Summary Generation

Summarization can typically perform the three primary summary forms via conclusion generators. These are Extraction-based, Abstraction-based, and Aided summarization. The extraction-based summarizing approach has been utilized to create the summary. With the Extraction-based type, the article summarizer will take small passages from the entire text and ultimately retain them. Finding phrases or specific words that could be utilized as tags for the text or document is the goal of the process, which is quite similar to extracting key phrases. An extraction-based summary generator gathers the necessary details from the original text and assembles them into a summary. When a student uploads a lecture or paragraph with raw text content, the system will identify the 'stop words' (such as am, is, are, pronouns, etc.) and remove them from the section. After that system identifies the most used words in the paragraph, and it will select the sentences with those words. Once the system identifies the sentences that include the most used words in the paragraph, it will be considered the most critical section. Then it will generate a summary from the paragraph as shown in Fig 2.

Polymers are made up of large molecules arranged in the form of long chains. The process of forming polymers is called polymerization. Large molecules formed by the joining of a large number of small molecules with one another are known as polymers. Pay attention to the chain formed by joining some paper clips together.

Fig. 2: Summarized Content

The above Feature is developed using Python language and Natural Language Processing (NLP). And the main libraries involved are Spacy and Heapq. Each parent node in a heap queue is less than or equal to its child nodes. It is implemented using Heapq module.

F. Generation of the Mind-Map

Once the system generates the paragraph summary of the uploaded lecture or paragraph, it will convert that summary to a mind map. The Topic Modeling approach has been used to develop this feature. Initially, the system extracts the topics of the auto-generated summary from this method. For instance, the system will extract every subtopic when a student provides a lesson or paragraph. After that, Topic Modeling will rearrange it to the importance of the subtopics. And those topics will be visualized on the front end as a mind map using React JS. By grouping the documents into categories, topic modeling is an unsupervised method that aims to analyze massive amounts of text data. For topic modeling, there are no labels associated with the text data. Instead, topic modeling looks to organize the documents into clusters based on shared traits. The main algorithms are Latent Dirichlet Allocation, Latent Semantic Analysis, and Correlated Topic Modeling to aid in the development of this functionality as shown in Fig 3.

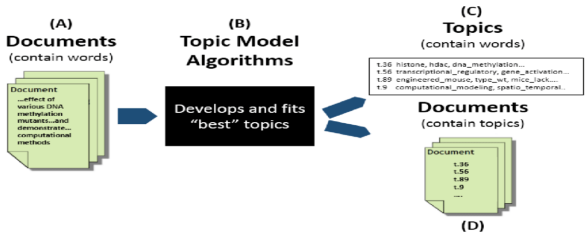


Fig. 3: Topic Modeling

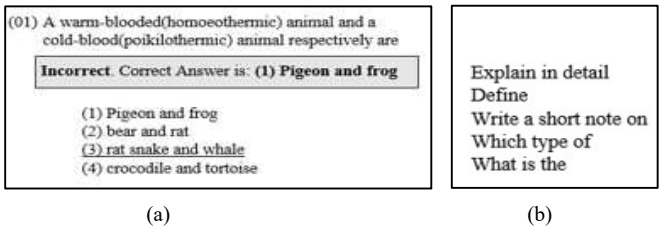
G. Personalized Guidance with MCQ Question evaluation

The past paper is provided to the student on the system's front end, which will record students' responses in a database. Question answers are stored separately, and it compares students' answers. The system automatically evaluates answers and displays if the answer is correct or not. If the answer is incorrect, the system displays the right answer. G.C.E Ordinary Level Science, a new syllabus (2016 – 2021), past papers with answer sheets, and grade 10 11 science textbooks are taken from e-thaksalawa. Students' responses will be analyzed by content reading with Pypdf and categorized into six levels by score, as shown in Table I.

TABLE I. CATEGORIZED LEVELS

Score	Level
0 - 35	W
35 - 45	F
45 - 55	S
55 - 65	C
65 - 75	B
75 - 100	A

Those score levels are customizable as per user requirements. The system generates questions using pre-defined question patterns with keywords. After that, the model transforms sentences into questions by omitting the answer and replacing it with a blank space. The optimum distractions for a given keyword are neither too similar nor too different and mutually exclusive, as determined by word embedding and cosine similarity. Similar questions will be generated as a supplementary plan by the student's level, identified by examining the student's responses. After that, the system will review students' responses repeatedly to monitor their progress. Python Natural Language Tool Kit (NLTK) is used to convert the chapter's content to machine language. Compared to NLTK to spaCy, NLTK has enjoyed dominance being an NLP standard in Python. Then spaCy disrupted this and found its own space. Finally, based on the score similar questions and answers will be generated as shown in Fig 4.



(a) (b)

Fig. 4: Question Generation Procedure: (a) Answer Evaluation: (b) Predefined Question Patterns

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IV. RESULT AND DISCUSSION

A. Object Detection API and SSD MobileNet

The major goal of the developed system is to create an artificial intelligence-powered system that can recognize students' emotions, external distractions, and facial expressions while they are learning. The system will display information regarding focus levels that will assist students in maintaining the best degree of attention to their academic studies. The developed system is capable of detecting four main emotions. Those will be happy, neutral, drowsy, and bored. The training dataset consists of more than 600 images in total, After the 10000 steps in the training process total _loss is 0.17, classification_loss is 0.05, localization_loss is 0.06, and regularization loss is 0.11 for the generated model. Additionally, by using the same system, instructors and other parties may also access the same data and information. The web camera will capture the student's face and send it to the SSD MobileNet V2 FPNLite 320x320 with a speed of 22ms and 22.2 COCO map frame by frame then the model will predict the emotion level and emotion sequentially as presented in Fig 5.

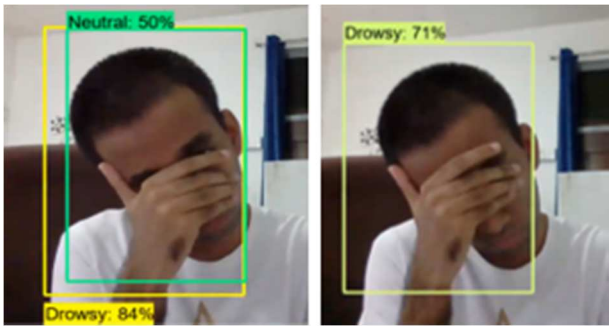


Fig. 5: Focus detection

After the development of the model, a testing stage was conducted to get the accuracy level of different facial expressions and emotion levels. Test results performed an overall confidence level of 85% for the model as illustrated in Table II.

TABLE II. FOCUS DETECTION TEST RESULTS

	Confidence rate of predicted levels				
Test states	***Actual emotion***	***Neutral***	***Happy***	***Drowsy***	***bored***
Test:01	***Drowsy***	***10%***	***5%***	***71%***	***50%***
Test:02	***Drowsy***	***12%***	***10%***	***70%***	***40%***
Test:03	***Bored***	***20%***	***10%***	***60%***	***82%***
Test:04	***Happy***	***50%***	***76%***	***20%***	***10%***

Compared to other systems, the developed system enhances the privacy level of users by providing a real-time focus level without storing any private information. With highly accurate speed and confidence rate, the model can identify the emotion levels periodically for the time frame. Using SSD MobileNet, the system can identify the most accurate predictions for students' facial expressions and emotions.

B. Essay-Based Question Evaluation

The technique for assessing essay responses used DistilBERT model that has been enhanced and can evaluate the degree of semantic similarity between two long texts. It was tuned using a 4:1 training and test split on the STSbentchmark dataset. The model's effectiveness with the testing dataset was evaluated using spearman's correlation and pearson correlation coefficient. In the fifth test, the DistilBERT model was improved to obtain a cosine similarity accuracy of 86.0%, while the accuracy of human (educators) evaluation is 84%, as shown in Table III.

TABLE III. ACCURACY LEVEL OF STUDENTS' ANSWERS

Test no		**Accuracy**	
	Question	***Evaluator***	***Proposed System***
1	Define blood Circulation	80%	78%
2	Structure and Function of the Heart	79%	81%
3	Explain the human digestive system	82%	81%
4	What is DNA?	84%	86%
5	What is a cell theory?	86%	84%

The system will next compare the student's response to the exact response on the answer sheet to evaluate its accuracy. Then it will extract the points for the response from the correct response that is stated in the answer sheet. Finally, the topics missed by the student will be displayed by the system.

C. Generation of the Summary

This model provides the ability to automatically create mind maps and summaries using NLP techniques with an overall accuracy rate of 80%. Only for testing purposes, educates evaluates, and provides an overall accuracy rate of 80% for the auto-generated summary from the system. Additionally, as demonstrated in Table IV, the test results system provides a comparable overall score of 70% for both the student-written summary and the computer-generated summary.

TABLE IV. ACCURACY LEVEL OF SUMMARY

Content	**Word count of the student-written summary**	**Word count of the auto-generated summary**	**Accuracy**
Paragraph - 01	63	72	61%
Paragraph - 02	93	78	65%
Paragraph - 03	51	55	88%

D. MCQ Question Evaluation

The model recognized the correct word sense in the sentence from a chapter paragraph and produced the question and answer. This demonstrates that the system could create questions based on given sentences and offer more generalized responses. The intent was to accurately recognize the keywords and make questions by omitting one of those keywords from the passage. The algorithm also classifies students for different levels based on how well they answered specific questions. According to Fig 6 automatic generated questions will generate more accurate questions along with the answers. Also, the system could generate direct short answers and essay-type questions. It also automatically evaluates question answers by displaying the correct answers. As a result, Students can enhance their academic performance

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by practicing with previous exams and questions of a similar nature.

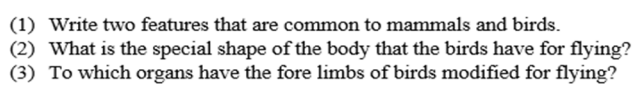


Fig. 6: Automatically generated Questions

V. CONCLUSION

This research demonstrates an approach to AI learning and analytics, which identifies the students' focus level, essay-based evaluation, mind map generation for lecture studies, and regular academic performance evaluation. This strategy brings up a fresh way to effectively bridge the gap between physical and distance learning productively. The study has colonized success by enhancing the various mechanisms that consistently support students to keep productive momentum. The strategies used in E-tutor are self-evaluation, encouraging to eliminate the unproductive stages, providing question papers, evaluating papers, generating newer questions, categorizing the users based on performance, visualization of the lecture content, and summarizing the lectures. Compared to other existing object detection models, E-tutor has custom object detection capable of detecting face expressions with high accuracy for a minimum train dataset. Also, rather than act as a single facility in other e-learning platforms, E-tutor combinations all other components, such as essay-based answer evaluation, note summarization, mind map generation, and personalized guidance facilities, to enhance the user's performance regularly. In future development, E-tutor mainly prioritizes strengthening the model's accuracy level by applying different datasets and enlarging the dataset. Furthermore, the developed system will expand the evaluation scope level of the subject and include other subjects to enhance the users' engagement.

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